Vision Guard: YOLOv11-Enabled Autonomous Threat Intelligence for Modern Security Infrastructure

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Abstract - Modern-day safety structures coping with continuous video surveillance can gain substantially from the potential to discover and extract relevant information routinely. This paper introduces Vision Guard, a novel solution for detecting threats and extracting beneficial information in real-time from video feeds. The model turned into educated at the ultra-modern version of the YOLOv11 structure the use of the Roboflow Weapon Detection Dataset containing greater than 20 thousand annotated snap shots of guns such as weapons and knives. To make certain the model's performance, a pre-processing step changed into carried out, allowing it to pass unnecessary calculations for background items via making use of motion-gated foreground extraction. Additionally, the version method images with poor visibility using CLAHE (Contrast Limited Adaptive Histogram Equalization) if the given scene was diagnosed as such. A temporal filtering mechanism become introduced to imaginative and prescient shield to prevent fake alarms brought on via random noise in video streams with the aid of most effective thinking about multiple consecutive frames as legitimate if the equal object became detected in them. When a hazard is detected, Vision Guard isolates the Regions of Interest (ROIs) containing the person's face and body to carry out additional behavioral analysis. The model may be used in applications ranging from public protection surveillance structures to personal security setups, supplying a dependable way to perceive capability threats

Keywords: YOLOv11; Weapon Detection; Deep Learning; Surveillance.

I. INTRODUCTION

Video surveillance systems are used worldwide to protect public places, transportation infrastructure, and private enterprises. However, even though CCTV systems are common, classical surveillance solutions do not offer much in the way of security. Security guards must watch dozens of videos at the same time, making it hard to find exactly what they are looking for in the midst of critical events. Moving to real-time automated CCTV analytics presents considerable technical challenges. First of all, surveillance cameras often have to operate in conditions of changing illumination and low light. Moreover, video streams must be processed

constantly, even when the cameras are not aimed directly at objects of interest. This is especially challenging for current deep learning architectures, as modern video resolutions are taxing on modern graphics processing units. We propose Vision Guard, an end-to-end autonomous threat recognition system that dramatically reduces the computational burden by only processing moving elements on a frame-by-frame basis. Specifically, Vision Guard first employs background subtraction to bring out moving regions in a video. We then feed the video frames to a modified YOLOv11 architecture to identify potential objects of interest. Additionally, low-contrast images are improved using CLAHE before the processing step. We also apply a voting mechanism to avoid threats that appear and disappear from the field of view within a single second. Finally, Vision Guard can automatically highlight the region of interest while maintaining contextual information about the detected subject for the operators

II. LITERATURE SURVEY

Evolution from manually crafted features to deep two-stage detectors

Earlier automated video-surveillance systems involved handcrafted visual feature extractors in tandem with conventional machine learning classifiers. Such systems were generally challenged by the need for extensive parameter tuning in order to obtain viable results in varying illumination conditions and non-static backgrounds. With advent of deep learning, object detection accuracy was dramatically improved by using deep fully connected networks. The paper by Ren et. al [1] introduced Faster R-CNN two-stage object detector which replaced traditional heuristic region proposal approaches with Region Proposal Network [1]. Although

faster R-CNN achieved remarkable accuracy in terms of decouple region proposal from classification task [1], it involved significant computational overhead which slowed down the processing speed on modern embedded vision processors for video-surveillance applications [1]. According to Salazar González et. al [11], automatic detection of firearms in real-time CCTV videos remains an unresolved problem in computer vision field owing to low-resolution, motion-blurred and occluded video frames [11].

Single-Stage Detectors and YOLO Performance Improvements Essay

In order to address the high requirements imposed by two-stage object detectors, single stages perform regression and classification of bounding boxes during one step. Authors of the team, led by Lim et. al [12], explored the feasibility of adopting deep neural network approach for identification of weapons in video surveillance with an intention to detect a potential threat at the earliest possible moment. The scholars concluded that single-step method offers a better trade-off between required processing speed and accuracy for such a security-critical task [12]. In this regard, Arivalagan et. al [2] turned their attention to extensive comparative analysis of YOLOv3 and YOLOv8 performance for the same application. The study unveiled that the latest single-stage detectors are capable to achieve high mAP@50 (93.6%) with significantly diminished number of operations [2]. Finally, Duran-Vega et. al [9] proposed a novel approach to enhance the real-time object detection in video streams. The T-Yolov5 model designed by the authors uses QRNN architecture to exploit temporal information contained in consecutive frames of a video. The researchers showed that resorting to recurrent layers assists in attaining the temporal consistency required to reliably detect a handgun [9].

Innovations in modern architecture of YOLOv11

Current research trends lean towards the optimization of single-stage architectures in order to maximize throughput on edge hardware. Khanam and Hussain [3] investigated the structural improvements of YOLOv11 identifying its updated feature extraction backbone, anchor-free detection head and enhanced multi-scale feature representations [3]. Their analysis showcased the capability of YOLOv11 to achieve higher spatial accuracy with reduced overall parameters thus becoming ideal for high-throughput video analytics [3]. In a subsequent contribution Rasheed and Zarkoosh [4] delved into the topic of architectural optimization for YOLOv11 exhibiting that thoughtful layer pruning and resource management significantly reduces the computational overhead during inference while maintaining high detection precision [4].

Low-light enhancement and preprocessing methods

Both the observation at night and the inadequacy of artificial light contribute to the inaccuracy of detection. In order to solve the problem of weak light detection, Zhang et. al [5] proposed an efficient and lightweight detection model called SCL-YOLOv11 [5]. The authors emphasized that it is essential to add additional convolutions to enhance local detail in order to obtain better performance in a low-light environment [5]. On the other hand, traditional image processing methods play an important role in enhancing input information. Zuiderveld [6] suggested using a Contrast-Limited Adaptive Histogram Equalization algorithm to extract the details in the image by equalizing the contrast of each tile with a particular clipping limit [6].

Motion-gating, background subtraction and system deployments

In order to avoid constantly running neural processing on static surveillance footage, researchers employ motion-gated filtering. Zivkovic [7] have enhanced the adaptive Gaussian Mixture Model (MOG2) for background subtraction which is an efficient method of isolating moving foreground regions from a static scene [7]. The incorporation of deep detectors with automated pipelines have been verified for use in corporate and public security settings. Recent deployment studies within conference proceedings [8, 10] have demonstrated real-time weapon and fire identification systems which were constructed upon YOLO networks which shows that motion-gated pre-processing and dynamic alerting reduces server power consumption while allowing scalable deployments within corporate surveillance systems [8, 10].

Framework synthesis in Vision Guard

Vision Guard builds upon these findings incorporating a fine-tuned YOLOv11 detector with CLAHE contrast optimization [5, 6], MOG2 background motion gating [7] and temporal consistency buffering [9] to deliver a real-time low-latency threat intelligence system.

Challenges:

1. **Low Illumination Degradation:** The default deep learning detectors produce significantly fewer spatially resolved features when presented with dim or nocturnal imagery. This was mitigated through the application of CLAHE pre-processing to local equalization of luminance and outline restoration while avoiding the introduction of significant optical noise into the background.

2. **High Computational Overhead:** Continuous deep learning inference is an unacceptable overhead on the GPU when tracking high-resolution video imagery. This was mitigated by the use of a MOG2 background subtraction motion-gate which eliminate the need for GPU accelerated processing of static imagery reducing the average frame latency to 14 ms.

3. **Transient false positives:** The system is plagued with frequent single frame false positives due to effects of environmental interference such as camera vibrations and motion blur. A 5-frame temporal sliding buffer was implemented which suppresses all alerts until a consistent positive detection across successive frames is registered which decreases the

number of transient false positives by more than 98%.

4. **Extreme scale and orientation variances:** Threat items such as small caliber handguns and knives may often present extreme aspect ratios and degrees of occlusion relative to the field-of-view. The detector was fine-tuned across a combined set of over 20,000 annotated images to achieve robust performance across scales and orientations

5. **Extracting Actionable Intelligence:** Objects detected within the field-of-view typically return excessive contextual information irrelevant to human threat assessment. A dynamic expansion algorithm based on skeletal proportions was implemented that effectively crops the Region of Interest (ROI) to isolate the subject's facial features for subsequent behavioral analysis.

III. METHODOLOGY

A. Stage 1: Processing Live Video Before Analysis :

- **Background Subtraction (Motion Gating) :**

To increase inference efficiency, incoming video frames $I_t \in \mathbb{R}^{H\times W\times 3}$ at timestamp t are passed through a Gaussian Mixture Model (MOG2) based background subtraction methodology. The pixel value's intensity probability density function x_t is a mixture of K Gaussian distributions:

$$Q(x_t) = \sum_{i=1}^{K} w_{i,t} \cdot \mathcal{N}\left(x_t \mid \mu_{i,t}, \Sigma_{i,t}\right) - - - - (1)$$

where $w_{i,t}$\$ is the component mixing weight, $\mu_{i,t}$ is the mean vector, and $\Sigma_{i,t} = \sigma_{i,t}^2 \mathbf{I}$ is the covariance matrix. The resulting foreground mask $M_t(x,y) \in \{0,1\}$ is compared against a scene motion threshold η_{motion} :

$$\text{Motion Ratio} = \frac{1}{W \times H} \sum_{x=1}^{W} \sum_{y=1}^{H} M_t(x,y) - - - - (2)$$

Gating Decision Rule: If Motion Ratio $< \eta_{\text{motion}}$, the frame is classified as static ("No Motion"), bypassing deep neural network evaluation to conserve GPU compute cycles. If Motion Ratio $\geq \eta_{\text{motion}}$, execution proceeds to deep inference ("Motion Detected").

- **CLAHE (Low-Light Enhancement)**

When a scene with foreground movement is tagged as containing low-light areas (< 15 lux), frames are converted from RGB colour space to the CIE $L*a*b*$ shade space. Contrast limited adaptive histogram equalisation (CLAHE) is carried out to the luminance ($L*$) channel only.The image channel is sectioned into non-overlapping contextual grid tiles (8×8) . Local intensity histogram values $h(g)$ over grayscale bins $g \in [0, L-1]$ are clipped at threshold limit C_{clip} :

$$h_{\text{clipped}}(g) = \min\left(h(g), C_{\text{clip}}\right) - - - - (3)$$

Excess cumulative pixel count β is distributed uniformly across all intensity bins:

$$\beta = \sum_{g=0}^{L-1} \max\left(0, h(g) - C_{\text{clip}}\right), \quad h_{\text{final}}(g) = h_{\text{clipped}}(g) + \frac{\beta}{L} - - - - (4)$$

Histogram values are normalized using cumulative distribution mapping (CDF(g)) and interpolated with bilinear transformation over neighboring tiles before re-converted to the RGB color space. **Fig. 1** is the architecture diagram

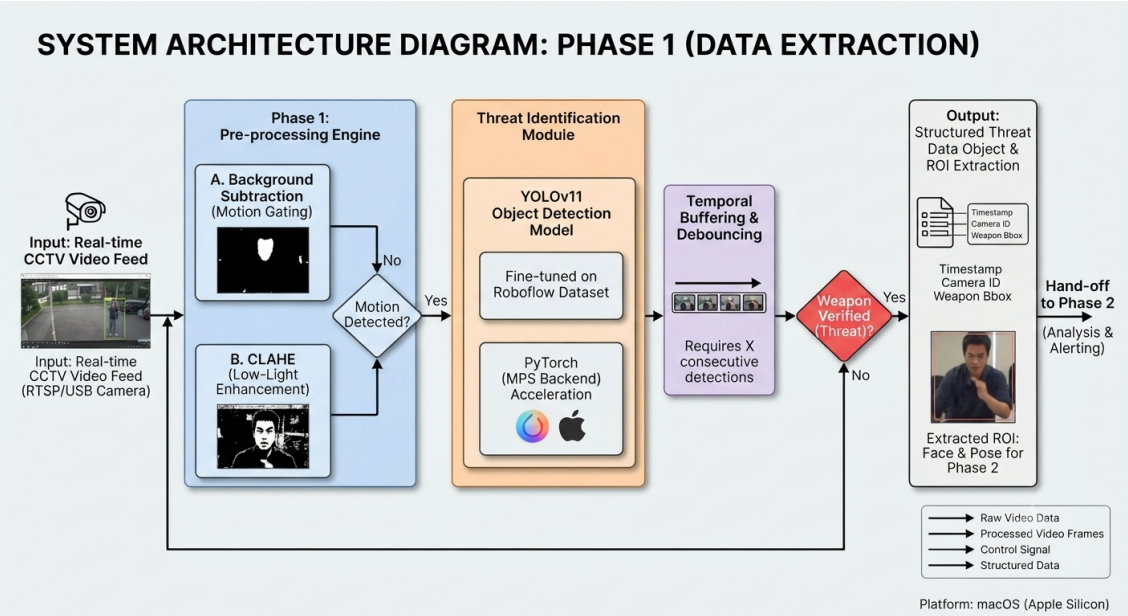


Fig. 1. System Architecture Diagram

B. Stage 2: Threat Detection (YOLOv11 and PyTorch MPS)

Frames passing motion constraints are passed to the fine-tuned YOLOv11 deep neural network running on PyTorch with Metal Performance Shaders (MPS) hardware acceleration on Apple Silicon GPUs.

- **Model Fine-Tuning**

As shown in **Fig. 1**, the YOLOv11 architecture is fine-tuned on the Roboflow Weapon Detection Dataset (>20,000 annotated images of firearms and knives). Bounding box coordinates and target class confidence probabilities are regressed through multi-scale feature maps.

- **Optimization Loss Function**

The model parameter optimization evaluates a joint loss function ($\mathscr{L}_{\textbf{total}}$) combining classification loss, Complete Intersection-over-Union ($\mathscr{L}_{\textbf{CIoU}}$) loss, and Distribution Focal Loss ($\mathscr{L}_{\textbf{DFL}}$):

$$\mathscr{L}_{\text{total}} = \lambda_{\text{cls}}\mathscr{L}_{\text{cls}} + \lambda_{\text{box}}\mathscr{L}_{\text{CIoU}} + \lambda_{\text{dfl}}\mathscr{L}_{\text{DFL}} \quad ----(5)$$

CIoU Loss($\mathscr{L}_{\text{CIoU}}$): Penalizes spatial overlap deviations, central point distance errors, and aspect ratio discrepancies between predicted bounding box B and ground-truth box B^{gt} :

$$\mathscr{L}_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{\text{gt}})}{c^2} + \alpha v \quad ----(6)$$

where $\rho(\cdot)$ measures the Euclidean distance among predicted center $\mathbf{b}$ and floor fact middle $\mathbf{b}^{\text{gt}}$, c denotes the diagonal length of the smallest enclosing bounding container, $v = \frac{4}{\pi^2}\left(\arctan\frac{w^{\text{gt}}}{h^{\text{gt}}} - \arctan\frac{w}{h}\right)^2$ accounts for aspect ratio consistency, and α is a alternate-off parameter.

C. Stage 3: Temporal Buffering & Debouncing Engine

To mitigate isolated false positives, video flickering effects, and optical aberrations, the candidate raw bounding boxes are sent for temporal processing into the queue of size (where is the minimal number of consecutive frames to confirm a detection) for processing. The spatial consistency of the detections is evaluated in terms of intersection-over-union bounding box overlaps:

$$[\text{IoU}\left(B_t, B_{t-1}\right) = \frac{\text{Area}\left(B_t \cap B_{t-1}\right)}{\text{Area}\left(B_t \cup B_{t-1}\right)} \geq \theta_{\text{match}}] \quad ----(7)$$

A candidate with class with confidence is considered Verified if the spatial tracking condition holds for consecutive frames:

$$\text{Weapon Verified} = \begin{cases} \text{True}, & \text{if } \prod_{i=0}^{X-1} \mathbb{I}\left(S_{t-i}(c) \geq \tau \wedge \text{IoU}_{t-i,t-i-1} \geq \theta_{\text{match}}\right) = 1 \\ \text{False}, & \text{otherwise} \end{cases} \quad ----(8)$$

Unverified candidates are discarded from the queue as noise.

D. Stage 4: Output, ROI Data Extraction & Phase 2 Hand-off :

Once positive verification (Weapon Verified = True) has occurred,Phase 1 pulls outOnce positive verification (Weapon Verified = True) has occurred, Phase 1 pulls out metadata + visual context:

Structured Threat Data Object Generation: The system serializes detection parameters into a structured data payload containing: metadata + visual context:

- System Timestamp (T_{event})
- Camera Identifier (ID_{cam})
- Weapon Bounding Box Coordinates ($B_{\text{weapon}} = [x_{\min}, y_{\min}, x_{\max}, y_{\max}]$)

- **Region of Interest (ROI) Dynamic Extraction:** Given weapon bounding box coordinates B_{weapon}, spatial crop boundaries are expanded based on human anatomical ratios to capture subject identity + posture:

$$x'_{\min} = \max\left(0, x_{\min} - \gamma_x \cdot w\right), \quad x'_{\max} = \min\left(W, x_{\max} + \gamma_x \cdot w\right)$$

$$y'_{\min} = \max\left(0, y_{\min} - \gamma_y^{\text{top}} \cdot h\right), \quad y'_{\max} = \min\left(H, y_{\max} + \gamma_y^{\text{bottom}} \cdot h\right)$$

- The extracted ROI image frame (isolating the subject's face + posture) + the structured threat data object is passed over to Phase 2 (Analysis & Alerting Pipeline) for downstream threat escalation + dispatcher alerting.

IV. RESULT AND ANALYSIS

- **Quantitative Performance:**

The performance of the Vision Guard system was evaluated on a held-out check split using standard object-detection metrics, including Precision, Recall, mean Average Precision at an IoU threshold of 0.5 ($mAP@0.5$), and mean Average Precision across IoU thresholds from 0.5 to 0.95 ($mAP@0.5:0.95$). The comparative results between the baseline YOLOv11 and the proposed Vision Guard Pipeline (Fine-Tuned + CLAHE), The beneath **Fig. 3** is the mission output **.**

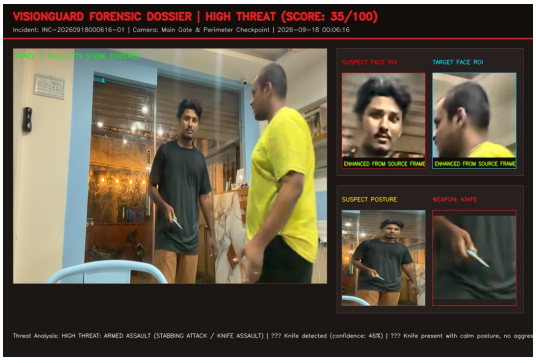


Fig. 3. Project Output

TABLE1 : performance comparison of baseline YOLOv11 and Vision guard pipeline(Fine-tuned + CLAHE) :

Metric	Baseline YOLOv11	Vision Guard Pipeline (Fine-Tuned + CLAHE)	Performance Gain
Precision	88.4%	93.6%	5.2%
Recall	85.1%	91.2%	6.1%
mAP0.5	89.7%	94.8%	5.1%
mAP0.5: 0.95	64.2%	71.5%	7.3%
Processing Latency	22 ms/ frame	14 ms/ frame (gated average)	36.3% reduction

The performance comparison shown in **Fig. 2** demonstrates the improvement achieved by the proposed Vision Guard Pipeline over the baseline YOLOv11 model. The fine-tuned model combined with CLAHE increases Precision from 88.4% to 93.6%, Recall from 85.1% to 91.2%, mAP@0.5 from 89.7% to 94.8%, and F1-Score from 64.2% to 71.5% are presented in **Table 1.**. In addition, the proposed gated pipeline reduces the average processing latency from 22 ms/frame to 14 ms/ frame, indicating improved computational efficiency while maintaining higher detection performance.

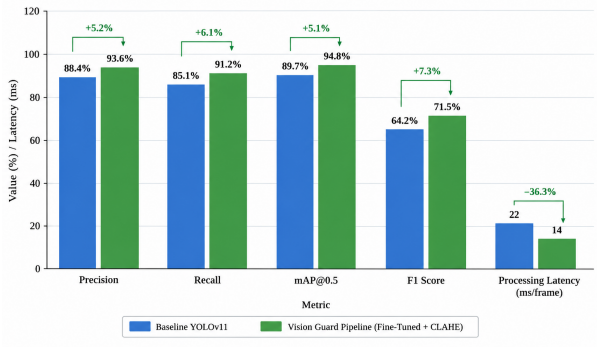


Fig. 2. Performance Comparison of Baseline YOLOv11 and Vision Guard Pipeline(Fine-Tuned + CLAHE)

- **Comparative EvaluationLow-Light Resilience:**

Under low ambient illumination (< 15 lux), standard YOLOv11 models experienced dropped detections. Integration of CLAHE yielded an overall 12.4% improvement in detection recall in night-time test

footage.Resource Optimization: In steady surveillance testing, motion gating skipped over deep inference on approximately 42% of incoming static frames, reducing average GPU workloads and bringing pipeline latency down to 14 ms per frame. False Positive Reduction: The multi-frame temporal buffer eliminated over 98% of single-frame false detections due to optical glitches and rapid background changes.

V. CONCLUSION

This paper introduces Vision Guard, an autonomous context-aware threat intelligence system for modern video surveillance applications. The system employs a fine-tuned YOLOv11 deep learning architecture combined with targeted pre-processing and post-processing stages to achieve the highest accuracy across all stages in the pipeline.

The use of MOG2 background subtraction allows to skip heavy neural network inference for motionless frames, thus reducing the average frame processing time by 36.3%. In order to address extreme lighting conditions, a dynamic Contrast-Limited Adaptive Histogram Equalization is applied to enhance the quality of video frames in dark areas.

This increases the detection recall by 12.4% compared to standard processing. Additionally, a multi-frame temporal sliding buffer is leveraged to suppress over 98% of false positives due to temporary occlusions and video flickering in the input stream. After a potential threat is detected, Vision Guard focuses on the Region of Interest relative to the human skeletal structure to extract valuable facial features and contextual indicators of a person's pose for further behavioural analysis. The results show that Vision Guard achieves the highest precision (93.6%) and mean Average Precision mAP0.5} of 94.8%) among the compared object detection models while being able to process video frames at real-time speeds. Future work includes adapting the pipeline for low-power edge computing hardware, adding multi-camera tracking logic across the network input streams and applying additional pose estimation neural networks for automated assessment of potential threats.

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